

Programmable AC Power Source User Manual

APS-4000 Series

MATRIX TECHNOLOGY INC.

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SAFETY INSTRUCTION

This chapter contains important safety instructions that must be followed when operating or storing the AC power source. Read the following before operation to ensure your personal safety and maintain the instrument in optimal condition.

Safety Symbols

The following safety symbols may appear in this manual or on the instrument:

	WARNING	Identifies conditions or practices that could result in injury or loss of life.
	CAUTION	Identifies conditions or practices that could result in damage to the instrument or other property.
	DANGER	High voltage.
	ATTENTION	Refer to the manual.
		Protective conductor terminal.
		Earth ground.

Safety Guidelines

	CAUTION	- Before plugging into AC mains, verify that the output voltage is compatible with the load. It is recommended to disconnect the load before connecting to AC mains. - Do not use this instrument near water. - Do not operate or touch this instrument with wet hands. - Do not open the casing while the instrument is connected to AC mains. - Avoid touching the metal contacts of the output terminals. - Do not operate in environments containing corrosive gases, sulfuric acid mist, or metal-corroding substances. - Do not use in dusty or highly humid environments, as this may degrade reliability or cause instrument failure. - Install the instrument in a location free from vibration. - Install in an ambient temperature range of -20 to 60° C. Note: Operation may become unstable outside the range of 0 to 40° C.
AC power source		- AC Input Voltage: 110V / 220 V AC (single-phase), $\pm 10\%$, 47 – 63 Hz. - Connect the protective grounding conductor of the AC power cord to earth ground to avoid electrical shock.
	WARNING	
Fuse		- Ensure the correct type of fuse is installed before power-up. - Replace the AC fuse with the same type and rating as the original. - Disconnect the power cord before replacing the fuse. - Verify the cause of the blown fuse is corrected before replacing it.
	WARNING	

1. INSTALLATION GUIDELINES

This chapter introduces the procedures for product unpacking, inspection, pre-use preparation, and storage.

1.1 Unpacking and Examination

1. Unpack the AC power source and check the enclosed accessories.
2. The product is sealed in a packing carton protected by EPE foam. If there is any damage to the packing carton, inspect the AC power source for deformation, scratches, or panel damage.
3. In case of any damage, please inform us or your local distributor immediately. Our service center will repair or replace the unit. However, please do not return it without notifying us or your local distributor.

Packing list:

- AC power source: 1 unit
- Operation manual: 1 pc
- RS-232 cable: 1 pc (optional)

1.2 Preparation Before Use

Input Power

The input power requirements of the instrument are 110 V AC or 220 V AC $\pm 10\%$ (47 – 63 Hz), single-phase. Please use a proper input power source before turning on the AC power source, and use a fuse with the same specifications as the original. The fuse specification is indicated on the rear panel of the AC power source.



WARNING

Disconnect the power cord before replacing the fuse.



WARNING

Always ensure a good connection of the grounding wire and make sure it is connected to earth ground. The 3-pin power plug of the AC power source must be connected to a socket with a grounding wire. If an extension cord is used, make sure it is properly grounded. When the 3-pin power plug is plugged into a properly grounded socket, the AC power source is grounded to earth.

Operation Environment

Location: The installation site must be away from chemical deposits, explosive materials, sulfuric acid mist, or other substances that cause metal corrosion. The installation site should always be kept clean and dry, and free from dust, high humidity, and vibration.

Temperature: 0° C to 40° C

Relative Humidity: 10% to 90%

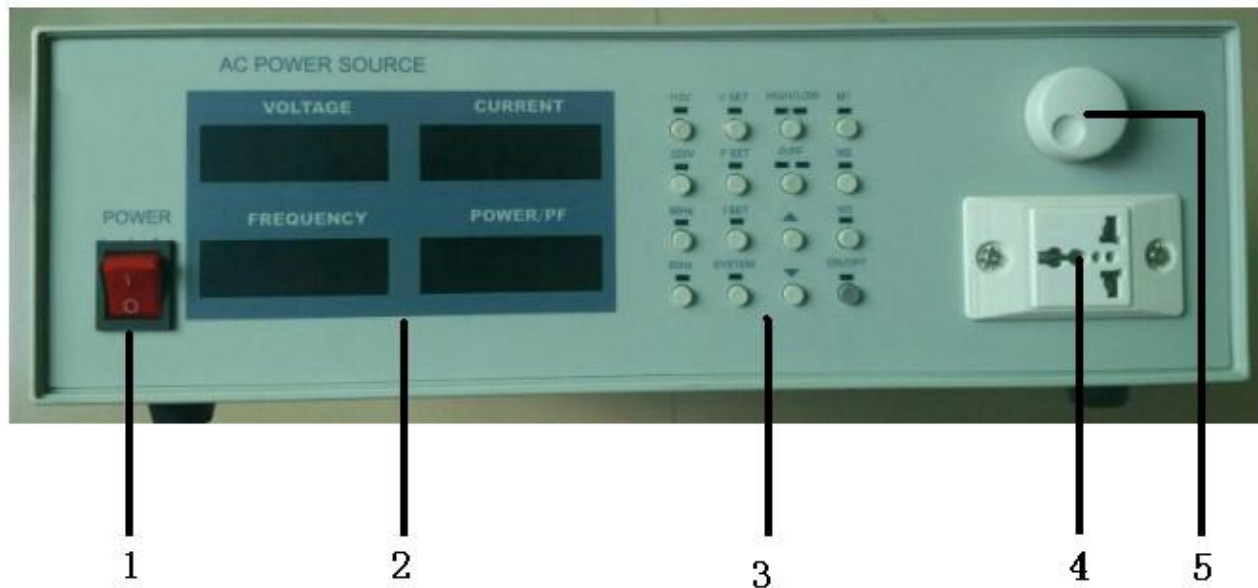
Altitude: <2000 feet

2. SPECIFICATIONS

Model		APS-4000A		APS-4000B		APS-4000C	
Capacity		350VA		700VA		1200VA	
Circuit mode		SPWM					
Input		Single phase 110 V AC or 220 V AC ±10% (factory-configured), 47-63Hz					
Output							
Phase		Single phase					
Voltage		Low range: 1.00 – 150.00 V / High range: 1.00 – 300.00 V					
Frequency		45 – 250 Hz, 0.1 Hz step					
Maximum Current	Low range	3A	6A		10A		
	High range	1.5A	3A		5A		
Load regulation		≤1%					
Line regulation		≤1%					
Harmonic distortion		≤2% (resistive load, L = 120 V, H = 240 V)					
Frequency stability		≤0.01%					
Crest factor		1.41 ± 0.10					
Efficiency		≥75%					
Response time		≤20 ms					
Frequency resolution		0.01 Hz					
Voltage resolution		0.01 V					
Current resolution		0.001 A					
Power resolution		4-digit LED, 0.001 W / 0.001 kW					
Power factor resolution		3-digit LED, 0.01					
Measuring accuracy	Voltage	±0.5% FS ±5 digits					
	Current	±0.5% FS ±5 digits					
	Frequency	±0.01% FS ±5 digits					
	Power	±0.5% FS ±5 digits					
Setting accuracy	Voltage	±1% FS					
	Frequency	±0.1% FS					
Communication interface		Optional RS-232 / RS-485 / USB					
Preset function		Preset output voltage and frequency					
Shortcut keys		110 V / 220 V, 50 Hz / 60 Hz					
Alarm function		Audible and visual alarm when protection is activated; displays error code on LED					
Protection		Fast-response circuit detects overcurrent, overload, overtemperature, and short circuit. Output shuts down in protection mode.					
Cooling		Fan-cooled					
Operation environment		0 – 40° C, ≤90% RH (non-condensing), Altitude: ≤2000 ft					
Dimension (W × H × D)		14.4 × 5.7 × 16.9 in					
Weight (lb)		28.0 lb	33.1 lb		40.8 lb		

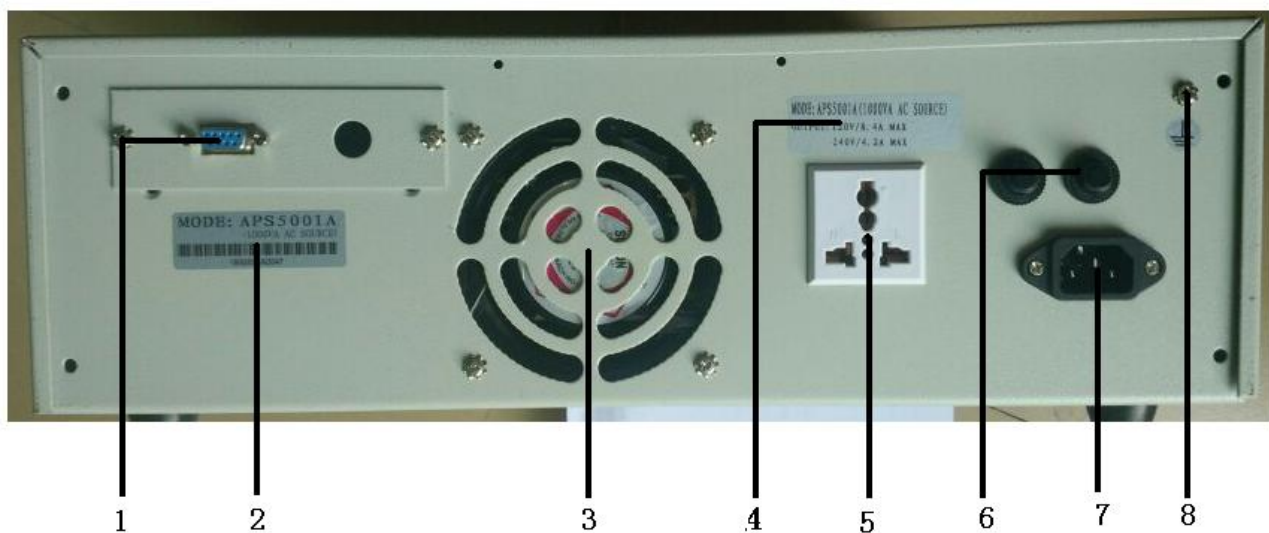
3. PANEL DESCRIPTION

3.1 Front Panel



1	Power switch	4	Output socket
2	Display	5	Rotary knob
3	Function keys		

3.2 Rear Panel



1	RS232, USB, RS485 interface (Optional)	5	Output socket
2	Serial number	6	Over current protector
3	Cooling fan	7	Input socket
4	Rating label	8	Ground terminal

3.3 Function Keys

Keys	Description
110V	Shortcut key for 110 V voltage output.
220V	Shortcut key for 220 V voltage output.
50Hz	Shortcut key for 50 Hz frequency output.
60Hz	Shortcut key for 60 Hz frequency output.
V Set	Press to enter voltage setting mode.
F Set	Press to enter frequency setting mode.
I Set	Press to enter current limit (maximum current) setting mode.
SYSTEM	Press to enter system configuration.
HIGH/LOW	Press to switch between high and low ranges. Default is high range; the indicator lights up.
P/PF	Press to toggle the display between power and power factor on the wattmeter. Default is power; the indicator lights up.
▲ Up key	Press to increase parameter (voltage/frequency/current) value.
▼ Down key	Press to decrease parameter (voltage/frequency/current) value.
M1, M2, M3 Memory keys	Short press to recall parameters; long press (>1 s) to store parameters.
ON/OFF	Press to turn the output on or off.

4. OPERATION INSTRUCTIONS



WARNING

- Before connecting to local AC mains, verify that the correct AC input voltage (110 V or 220 V) is used.
- To avoid electrical shock, connect the protective grounding conductor of the AC power cord to earth ground.
- Turn off the front panel power switch before connecting.
- Ensure 200 mm of clearance at the front and rear of the AC power source for proper ventilation.
- Do not use the AC power source during thunderstorms.

4.1 Voltage Setting

In standby or output mode, press the **V SET** key to enter voltage setting mode. Press the **▲** or **▼** key to increase or decrease the voltage value. There are four adjustment steps for the two keys: 0.01 V, 0.1 V, 1 V, and 10 V. Press the key once; the voltage changes by 0.01 V. If the key is pressed and held, the adjustment step changes from low to high every 0.3 seconds. The voltage changes faster and faster. If the new voltage

value remains unchanged for 2 s, the voltage meter flashes twice, and then the new voltage is output and displayed.

In voltage setting mode, the voltage can also be set using the rotary knob. Depending on the rotation speed, the adjustment step is 0.01 V, 0.1 V, or 1 V. The faster the knob is rotated, the larger the adjustment step (up to 1 V).

There are two output voltage ranges: low range (0 – 150 V) and high range (0 – 300 V). The default is the high range. The output voltage can be adjusted within the selected range.

4.2 Frequency Setting

In standby or output mode, press the **F SET** key to enter frequency setting mode. The output frequency can be set from 45 Hz to 250 Hz, with adjustment steps of 0.1 Hz, 1 Hz, and 10 Hz. Frequency adjustment is the same as voltage adjustment.

4.3 Current Limit Setting

In standby or output mode, press the **I SET** key to enter current limit setting mode. The maximum current limit varies by model; please refer to the specifications. The current limit adjustment step also varies by model: 0.001 A, 0.01 A, 0.1 A, and 1 A. Current limit adjustment is the same as voltage adjustment.

4.4 High/Low Voltage Range Switch

Press the **HIGH/LOW** key to switch between the high and low voltage ranges. The rated current in the high range is half of that in the low range. For rated current details, please refer to the specifications.

Do not switch the voltage range while the output is on. Doing so will shut down the output for approximately 20 ms and may cause unstable output.

There are two output voltage ranges: low range (0 – 150 V) and high range (0 – 300 V). The default is the high range. The output voltage can be adjusted within the selected range.

4.5 ON/OFF Key

Press the **ON/OFF** key to turn the output on or off. When the ON/OFF indicator lights up, the output is on. When the indicator is off, the output is off.

In output mode, if an error occurs, the output will shut down immediately. The output indicator will flash, and the beeper will sound. To silence the alarm, press the **ON/OFF** key. To reset, press the **ON/OFF** key again. To start the output, press the **ON/OFF** key once more.

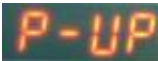
4.6 Memory Key

The AC power source can store three sets of voltage and frequency settings. There is one key for each memory location: M1, M2, and M3. After setting the voltage and frequency values, press one of the memory keys to store the settings. The stored settings can be recalled by short-pressing the corresponding memory key.

4.7 System Setup

In standby mode, press the **SYSTEM** key to enter system setup mode. Short-press the **SYSTEM** key repeatedly to cycle through the parameters in the following sequence: baud rate, panel lock, output commands, delay time, and address.

When a parameter is displayed, press the ▲ or ▼ key to change its setting.

Parameter	Display	Description
Baud Rate		Press the ▲ or ▼ key to change the baud rate to 2400, 4800, 9600, 19200, or 38400. Default: 9600
Panel lock		The panel lock can be set to ON or OFF by pressing the ▲ or ▼ key. When set to ON, the front panel keys are locked. Only the "ON/OFF" and "SYSTEM" keys can be used. Default: OFF
Output Status		The output status can be set to ON or OFF by pressing the ▲ or ▼ key. When set to ON, the AC power source provides output immediately after power-on. The front panel keys are locked. Default: OFF
Commands		The commands used for this AC power source can be selected by pressing the ▲ or ▼ key from three modes: "0": turn off remote communication; "1": use ASCII commands; "2": use HEX commands. Default: 1
		Reserved
Delay Time		The output delay time can be set by pressing the ▲ or ▼ key over a range of 0.001 s to 99.999 s. Default: 0.3 s
Address		The communication address can be set by pressing the ▲ or ▼ key over a range of 0 to 30. Default: 1

5. REMOTE CONTROL

5.1 Serial Port

The power supply is equipped with RS-232, USB, and RS-485 interfaces, supporting ASCII and HEX

commands.

5.2 Communication Commands

5.2.1 Commands to Query Parameters

Commands	Parameters	Description
?MAXPOW		To query maximum power.
?MAXVOL		To query maximum voltage.
?MAXCUR		To query maximum current.
?MAXFRE		To query maximum frequency.
?MINFRE		To query minimum frequency.

5.2.2 Commands to Set Parameters

Commands	Parameters	Description
PON		To turn on output.
POFF		To turn off output.
SVOL n		To set voltage.
SFRE n		To set frequency.
SCUR N		To set current limit.
?SVOL		To query the present voltage setting.
?SFRE		To query the present frequency setting.
?SCUR		To query the present current limit setting.
?MVOL		To query the present measured voltage.
?MFRE		To query the present measured frequency.
?MCUR		To query the present measured current.
?MPF		To query the present measured power factor.
?MPOW		To query the present measured power.

5.2.3 Complete Command Frame

1. Request Frame

Frame format: ID (1 byte) + function code (1 byte) + data (4 bytes) + check code (1 byte)

Command			Description
Instrument ID	1-28		ID of the AC power source.
Command	ASCII	HEX	
	'R'	0x52	Read data
	'W'	0x57	Write data
	'X'	0x58	Reset software
Function code	Refer to details in the next table		The parameter for the operation.
Data	High byte follows low byte.		Write data: write data to the AC power source.
			Read data: data reply to the PC.
			Reset command: ignored.
Check code			The accumulated sum of the first seven bytes.

Note: The AC power supply will not return any response after receiving the software reset command.

2. Function Code

Function Code	Function Description	Data Description		Meaning of Read Data	Meaning of Written Data
0x30	Output mode	Byte 0	Overcurrent status	1: Over current 0: Normal	0: Clear overcurrent flag
		Byte 1	Malfunction alarm	1: Malfunction 0: Normal	1: Clear alarm flag
		Byte 2	Voltage range (high/low)	1: High range 0: Low range	Ignored
		Byte 3	Output state	1: Output on 0: Output off	Ignored
0x31	Target frequency	Frequency, 4 bytes, 0.1 Hz resolution, range 45.0 – 250.0 Hz		Present frequency	Updated frequency
0x32	Target Voltage (High Range)	Voltage, 4 bytes, 0.1 V resolution, range 0 – 300.0 V		Present voltage	Updated voltage. If the previous voltage is in the low range, it switches to the high range
0x33	Target Voltage (Auto Range)	Voltage, 4 bytes, 0.1 V resolution, range 0 – 300.0 V		Present voltage	Updated voltage. The voltage range changes accordingly. If the voltage is above 150.0 V, it switches to the high range
0x34	Max. output current	Current, 4 bytes, unit: 0.001 A, range: up to 30000		Current limit value	Current limit value
0x35	Output on/off	Output status, 4 bytes (read only)	1: Output on 0: Output off.	Turn output on	
				Turn output off	
0x36	Output on/off	Output status, 4 bytes (read only)	1: Output on 0: Output off	Turn output off	
0x4A	Serial number	4 bytes		Serial number	Cannot write
0x60	Irms	RMS current, 4 bytes, 0.001 A resolution		RMS current value	Cannot write
0x61	Vrms	RMS voltage, 4 bytes, 0.1 V resolution		RMS voltage value	Cannot write
0x62	Ipeak	Peak current, 4 bytes, 0.001 A resolution		Peak current value	Cannot write
0x63	Vpeak	Peak voltage, 4 bytes, 0.1 V resolution		Peak voltage value	Cannot write
0x64	Pva	Apparent power, 4 bytes, 0.1 VA resolution		Apparent power value	Cannot write
0x65	Pw	Active power, 4 bytes, 0.1 W resolution		Active power value	Cannot write
0x66	Pf	Power factor, 4 bytes, 0.001 resolution		Power factor value	Cannot write
0x67	Freq	Frequency, 4 bytes, 0.1 Hz resolution		Frequency value	Cannot write

5.2.4 Command Frame Examples

The following command frame examples are given in HEX code, with the instrument ID set to 0x01.

- | | |
|--|--|
| 1) Set voltage of auto range to 120 V | 5) Set frequency to 60 Hz |
| Request frame: 01 57 33 B0 04 00 00 3F | Request frame: 01 57 31 58 02 00 00 E3 |
| Reply frame: 01 57 33 B0 04 00 00 3F | Reply frame: 01 57 31 58 02 00 00 E3 |
| 2) Set voltage of auto range to 240 V | 6) Set output ON |
| Request frame: 01 57 33 60 09 00 00 F4 | Request frame: 01 57 35 00 00 00 00 8D |
| Reply frame: 01 57 33 60 09 00 00 F4 | Reply frame: 01 57 35 01 00 00 00 8E |
| 3) Set voltage of high range to 120 V | 7) Set output OFF |
| Request frame: 01 57 32 B0 04 00 00 3E | Request frame: 01 57 36 00 00 00 00 8E |
| Reply frame: 01 57 32 B0 04 00 00 3E | Reply frame: 01 57 36 00 00 00 00 8D |
| 4) Set voltage of high range to 240 V | 8) Clear error |
| Request frame: 01 57 32 60 09 00 00 F3 | Request frame: 01 57 30 00 01 00 00 89 |
| Reply frame: 01 57 32 60 09 00 00 F3 | Reply frame: 01 57 30 00 00 00 00 88 |

Remarks:

- 1) The request frames for setting voltage and setting frequency must be sent before sending request frame to set the output ON. While the output is ON, if a new request frame for setting voltage or frequency is sent, the AC power source will respond immediately and update the output.
- 2) The AC power source turns off the output immediately once the request frame to set the output OFF is sent.
- 3) If the output is short-circuited or an error occurs, send the request frame to clear the error. The output will be turned off immediately..

6. MAINTENANCE

6.1 Inspection

- Inspect the AC power source at regular intervals so that it maintains its initial performance over time.
- Check the input power cord for damage to the vinyl cover and overheating of the plug and cord stopper. Check the terminal screws and binding posts for looseness.
- Remove dust from the inside of the casing and the ventilation holes of the cover using compressed air or the exhaust air of a vacuum cleaner.
- If the AC power source has not been used for a long time, turn it on and let it warm up for two hours every three months.

6.2 Cleaning

- Before cleaning, disconnect the AC power source from the mains.
- To clean the AC power source, use a soft cloth dampened in a solution of mild detergent and water. Do not spray cleaner directly onto the AC power source, as it may leak into the cabinet and cause damage.
- Do not use chemicals containing benzene, toluene, xylene, acetone, or similar solvents.
- Do not use abrasive cleaners on any portion of the AC power source.

6.3 Modification

- No modifications to the circuit or components may be made to the AC power source by the user.
- If the user makes any modifications, the warranty for the AC power source will automatically become void.
- We assume no responsibility for any consequences arising thereafter.
- If the returned unit is found to have any modifications, we will restore the AC power source to its original condition and charge for the maintenance service.

7. Warranty Card

What This Warranty Covers?

If the product fails due to a defect in materials or workmanship, MATRIX will provide free repair during the warranty period. If the product fails due to improper operation, misuse, or negligence, MATRIX will provide paid repair service within the warranty period.

How Long This Warranty Lasts?

This warranty is valid for three (3) years from the date of original purchase of all MATRIX-branded products.

Who Is Covered?

This warranty covers only the original purchaser of this product. It is not transferable to subsequent owners or purchasers.

How to Obtain Warranty Service?

If your product develops a problem, please contact your local distributor first. If you cannot locate a local distributor, you may contact us directly:

Email: service@szmatrix.com

Phone: +86 755 2836 4276

What information do customers need to supply?

Model No.	
Serial No.	
Description of the problem	
Photos	
Video (if necessary)	